Advanced Linear Algebra – Theoretical Depth & Relationships

1. Trace and Eigenvalues

For a square matrix A, the characteristic equation is $|A - \lambda I| = 0$.
For a 2×2 matrix, the characteristic equation expands to:
$\lambda^2 - S1\lambda + S2 = 0$,
where S1 is the sum of diagonal elements (Trace) and S2 is the determinant.

If $\lambda 1$ and $\lambda 2$ are eigenvalues, then:
$\lambda 1 + \lambda 2 = \text{Trace}(A)$.

For a 3×3 matrix:
$\lambda^3 - S1\lambda^2 + S2\lambda - S3 = 0$,
where S1 equals the trace and S3 equals determinant.
Thus, Trace(A) equals the sum of eigenvalues.

2. Symmetric Matrix Theorems

If A = A■ (symmetric matrix):

- All eigenvalues are real.
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- A can be diagonalized as A = Q D Q■.

Proof Idea:
If $Av1 = \lambda 1 v1$ and $Av2 = \lambda 2 v2$,
then $(\lambda 1 - \lambda 2)$(v1■v2) = 0.
If $\lambda 1 \neq \lambda 2$, then v1■v2 = 0.
Thus eigenvectors are orthogonal.

3. Matrix Decompositions

Eigen Decomposition:
If A has n independent eigenvectors:
A = P D P■¹,
where P contains eigenvectors and D contains eigenvalues.

For symmetric matrices:

A = Q D Q■.

Singular Value Decomposition (SVD):
For any m×n matrix:
A = U Σ V■,
where U and V are orthogonal and Σ contains singular values.

Eigen decomposition applies to square matrices.
SVD applies to all matrices.

4. Orthogonal Matrix Inverse

If Q is orthogonal:
Q■Q = I,
so Q■^{1} = Q■.

Orthogonal matrices preserve lengths, angles, and dot products.
They represent rotations and reflections.

5. Singularity and Rank

A matrix is singular if det(A) = 0.
Since determinant equals the product of eigenvalues,
if any eigenvalue is zero, the matrix is singular.

Rank properties:
If rank(A) = n → Unique solution.
If rank(A) < n → Infinite solutions.
Rank-nullity theorem:
rank(A) + nullity(A) = n.

6. Orthogonal Complement and Dimensionality

If W is a subspace of R■ with dimension k,
then its orthogonal complement W⊥ has dimension n – k.

Every vector v ∈ R■ can be written uniquely as:
v = w + w⊥,
where w ∈ W and w⊥ ∈ W⊥.

Projection formula:
$\text{proj}■(u) = (u \cdot a / ||a||^2)\, a$.